

TECHNICAL DOCUMENTATION: HEAT EXCHANGER SYSTEM

Synthetic Equipment Record

IDENTITY INFORMATION

Company_Name: Meridian Petroleum Operations LLC

Field_Engineer: James R. Hutchins

Field_Engineer_Id: MPO-ENG-4782

Document_Version: 2.3.1

Machine_Name: Thermal Management Unit - Platform Delta-7

Machine_Id: TMU-D7-2847

Installation_Date: 2022-03-15

Number_Of_Units_In_Machine: 3

UNIT 1: PRIMARY HEAT EXCHANGER CORE

Unit_Name: Shell-and-Tube Heat Exchanger Assembly (Primary)

Unit_Id: HX-P-001-D7

Manufacture_Date: 2021-11-22

Manufacturer_Company_Name: ThermoFlow Industrial Systems Inc.

Manufacturer_Company_Id: TIS-MFG-5634

Inventory_Quantity: 2 units in stock

Supplier_Information: ThermoFlow Industrial Systems Inc., Houston TX; Contact: procurement@thermoflow-ind.com; Lead Time: 8-10 weeks

DOMAIN SPECIFICATIONS:

- **Wellbore_Depth_Rating:** 12,000 ft equivalent thermal load
- **Operating_Pressure_PSI:** 450 PSI (shell side) / 380 PSI (tube side)
- **Drilling_Fluid_Compatibility:** Water-based and synthetic-based muds; mineral oil incompatible
- **Known_Error_Codes:** ERR-HX-101 (fouling detected), ERR-HX-203 (pressure differential exceeded)
- **Replacement_Part_Number:** TIS-HX-4500-SS-450

MAINTENANCE_SCHEDULE:

- Quarterly pressure differential monitoring
- Semi-annual tube bundle inspection
- Annual chemical cleaning cycle

- Biennial hydrostatic pressure test (API RP 54 Section 5.2)

SECTION 1 - USE CASE

- Removes excess thermal energy from drilling mud circulation system during high-temperature wellbore operations
- Maintains mud temperature between 85°F–120°F to preserve fluid properties and equipment integrity
- Supports extended reach wells (ERW) and deepwater operations requiring precise thermal control
- Integrates with mud cooling tower as secondary backup system
- Operates continuously during drilling phases; standby mode during tripping operations

SECTION 2 - FAILURE CAUSES

- **ERR-HX-101 (Fouling/Scale Buildup)** — Frequency: 45% of failures; caused by mineral precipitation in hard water systems; reduces heat transfer efficiency by 30–50%
- **ERR-HX-203 (Pressure Differential Exceedance)** — Frequency: 28%; tube-side blockage from drill solids or corrosion debris; triggers automatic bypass valve
- **Tube Corrosion/Pitting** — Frequency: 15%; galvanic corrosion in saltwater mud systems; perforation leads to cross-contamination
- **Gasket Degradation** — Frequency: 8%; elastomer breakdown from thermal cycling and chemical exposure; manifests as slow weeping leaks
- **Vibration-Induced Tube Erosion** — Frequency: 4%; high-velocity mud flow causes fretting wear at tube support plates

SECTION 3 - SOLUTIONS

- **Fouling Mitigation (ERR-HX-101):** Implement quarterly chemical cleaning using approved inhibited acid per API RP 54 Section 6.1; install 200-micron inlet strainer upstream; monitor differential pressure daily (threshold: 15 PSI)
- **Pressure Management (ERR-HX-203):** Install redundant pressure relief valve set at 420 PSI; perform monthly bypass valve functional test; maintain solids control equipment per API RP 13C standards
- **Corrosion Prevention:** Specify duplex stainless steel (2205) tube material for saltwater applications; apply zinc-based corrosion inhibitor to mud system; conduct annual ultrasonic thickness survey (OSHA 1910.119 Process Safety Management)
- **Gasket Replacement Protocol:** Replace all elastomer gaskets during scheduled maintenance; use EPDM or Viton per equipment rating; torque flange bolts to 85 ft-lbs in star pattern; pressure test at 1.5× operating pressure before return to service
- **Vibration Control:** Install elastomeric vibration isolators under unit feet; limit mud flow velocity to ≤8 ft/sec through exchanger; balance rotating equipment in mud system per ISO 20816 standards

SECTION 4 - INVENTORY DATA

- **Current Stock:** 2 complete primary heat exchanger assemblies; 8 replacement tube bundles; 12 gasket kits (EPDM); 4 pressure relief valve cartridges
- **Reorder Threshold:** Maintain minimum 1 complete unit + 4 tube bundles in inventory; trigger purchase order when stock reaches threshold
- **Supplier:** ThermoFlow Industrial Systems Inc. (primary); backup supplier: CryoTech Thermal Solutions (lead time 12 weeks)
- **Lead Time & Logistics:** Standard delivery 8–10 weeks; expedited air freight available at 15% premium; spare parts (gaskets, relief valves) stock 2–3 week lead time

UNIT 2: AUXILIARY HEAT EXCHANGER (BACKUP COOLING)

Unit_Name: Plate-Frame Heat Exchanger Assembly (Auxiliary)

Unit_Id: HX-A-002-D7

Manufacture_Date: 2021-12-08

Manufacturer_Company_Name: CompactTherm Engineering Ltd.

Manufacturer_Company_Id: CTE-MFG-7291

Inventory_Quantity: 1 unit in stock

Supplier_Information: CompactTherm Engineering Ltd., Aberdeen UK; Contact: sales@compacttherm-eng.co.uk; Lead Time: 10-12 weeks

DOMAIN SPECIFICATIONS:

- **Wellbore_Depth_Rating:** 8,000 ft equivalent thermal load
- **Operating_Pressure_PSI:** 300 PSI (both sides)
- **Drilling_Fluid_Compatibility:** Water-based muds only; not rated for oil-based systems
- **Known_Error_Codes:** ERR-HX-301 (plate separation), ERR-HX-402 (seal failure)
- **Replacement_Part_Number:** CTE-HX-2800-PF-300

MAINTENANCE_SCHEDULE:

- Monthly visual inspection for leakage
- Quarterly plate pack disassembly and cleaning
- Semi-annual seal replacement
- Annual pressure test at 450 PSI (1.5× operating pressure)

SECTION 1 - USE CASE

- Provides secondary cooling capacity during peak thermal loads or primary unit maintenance

- Compact footprint suitable for offshore platform space constraints
- Handles lower-viscosity water-based mud systems with enhanced heat transfer efficiency
- Operates in parallel with primary exchanger during extended drilling campaigns
- Automatic switchover via solenoid valve if primary unit pressure exceeds 420 PSI

SECTION 2 - FAILURE CAUSES

- **ERR-HX-301 (Plate Separation)** — Frequency: 52%; thermal expansion mismatch or improper bolt torque; allows mud bypass and loss of cooling capacity
- **ERR-HX-402 (Elastomer Seal Failure)** — Frequency: 35%; chemical attack from drilling additives or thermal degradation; manifests as cross-flow contamination
- **Plate Fouling** — Frequency: 10%; mineral deposits on narrow flow channels; reduces effectiveness by 40–60%
- **Bolt Loosening** — Frequency: 3%; vibration-induced fastener relaxation; detected during routine torque verification

SECTION 3 - SOLUTIONS

- **Plate Separation Prevention (ERR-HX-301):** Verify bolt torque to 120 ft-lbs per manufacturer spec every 500 operating hours; use thread-locking compound (Loctite 243) on all fasteners; monitor inlet/outlet temperature differential (target: 15–25°F); replace unit if differential drops below 8°F
- **Seal Management (ERR-HX-402):** Stock 6 replacement seal kits (EPDM) per unit; replace seals during quarterly disassembly; inspect for chemical compatibility with mud additives per MSDS review; conduct pressure decay test (≤ 2 PSI/hour acceptable per API RP 54)
- **Fouling Control:** Implement 150-micron inlet filtration; perform chemical cleaning every 250 operating hours using citric acid solution (pH 3.5–4.0); flush with fresh water post-cleaning; document cleaning intervals in maintenance log
- **Fastener Security:** Conduct bolt torque audit every 6 months using calibrated torque wrench; replace any corroded fasteners with duplex stainless steel (A4-70); apply anti-seize compound to threads
- **Operational Limits:** Restrict mud flow rate to ≤ 40 GPM through auxiliary unit; monitor pressure differential continuously; activate alarm if differential exceeds 20 PSI (fouling indicator)

SECTION 4 - INVENTORY DATA

- **Current Stock:** 1 complete auxiliary plate-frame exchanger; 6 replacement seal kits; 24 replacement plates (mixed types); 1 complete bolt/fastener kit
- **Reorder Threshold:** Maintain minimum 1 complete unit + 3 seal kits in inventory; order when stock reaches single unit

- **Supplier:** CompactTherm Engineering Ltd. (primary); secondary source: Hydro-Thermal Systems GmbH (Germany, 14-week lead time)
- **Lead Time & Logistics:** Standard 10–12 weeks; seal kits available from stock (2-day delivery); expedited unit delivery via air freight (3 weeks, 20% surcharge)

UNIT 3: COOLING WATER CIRCULATION PUMP (INTEGRAL COMPONENT)

Unit_Name: Centrifugal Cooling Water Pump Assembly

Unit_Id: HX-P-003-D7

Manufacture_Date: 2022-01-10

Manufacturer_Company_Name: FluidDynamics Pump Corporation

Manufacturer_Company_Id: FDP-MFG-3847

Inventory_Quantity: 1 unit in stock

Supplier_Information: FluidDynamics Pump Corporation, New Orleans LA; Contact: tech-support@fluidynamics-pump.com; Lead Time: 6-8 weeks

DOMAIN SPECIFICATIONS:

- **Wellbore_Depth_Rating:** N/A (surface equipment)
- **Operating_Pressure_PSI:** 85 PSI discharge (seawater cooling loop)
- **Drilling_Fluid_Compatibility:** Seawater/freshwater only; not in contact with drilling mud
- **Known_Error_Codes:** ERR-HX-501 (cavitation), ERR-HX-602 (bearing wear)
- **Replacement_Part_Number:** FDP-PUMP-1500-SS-85

MAINTENANCE_SCHEDULE:

- Weekly vibration monitoring (ISO 20816 limits)
- Monthly bearing temperature checks
- Quarterly seal inspection and lubrication
- Semi-annual impeller inspection for erosion
- Annual bearing replacement (planned maintenance)

SECTION 1 - USE CASE

- Circulates seawater through heat exchanger cooling jacket at 1,500 GPM nominal flow rate
- Maintains consistent cooling water supply pressure (80–90 PSI) to primary and auxiliary exchangers
- Operates continuously during drilling operations; shuts down during non-productive time
- Equipped with variable frequency drive (VFD) for flow optimization and energy efficiency
- Integrates with automated control system; alarm triggers if discharge pressure drops below 75 PSI

SECTION 2 - FAILURE CAUSES

- **ERR-HX-501 (Cavitation)** — Frequency: 38%; occurs when inlet pressure drops below vapor pressure; causes impeller pitting and noise; triggered by clogged intake strainer or low seawater level
- **ERR-HX-602 (Bearing Wear)** — Frequency: 42%; saltwater corrosion and inadequate lubrication; manifests as increased vibration and temperature rise; bearing life typically 12–18 months in marine service
- **Seal Leakage** — Frequency: 15%; mechanical seal degradation from thermal cycling; allows seawater ingress into bearing cavity
- **Impeller Erosion** — Frequency: 4%; sand/silt particles in seawater intake; reduces flow capacity by 10–20%
- **Motor Winding Failure** — Frequency: 1%; rare; caused by saltwater spray ingress or inadequate IP65 enclosure protection

SECTION 3 - SOLUTIONS

- **Cavitation Prevention (ERR-HX-501)**: Install 100-micron intake strainer with differential pressure gauge; clean strainer weekly; maintain minimum 5 ft static head on pump inlet; verify seawater level in cooling reservoir before each shift; set VFD minimum speed to prevent suction-side pressure drop below 10 PSI absolute
- **Bearing Management (ERR-HX-602)**: Implement planned bearing replacement every 12 months (or 8,000 operating hours) per OSHA 1910.119; use marine-grade grease (NLGI Grade 2, lithium complex); monitor bearing temperature weekly (alarm threshold: 180°F); replace seals during bearing change-out
- **Seal Integrity**: Inspect mechanical seal faces monthly for wear patterns; replace seal kit every 6 months or if leakage exceeds 5 drops/minute; use dual-cartridge seals rated for saltwater service; pressure test seal cavity at 100 PSI annually
- **Impeller Protection**: Install 200-micron strainer at seawater intake; flush intake line monthly with fresh water; inspect impeller for erosion during annual maintenance; replace if erosion depth exceeds 2 mm
- **Motor Protection**: Verify IP65 enclosure rating quarterly; apply silicone conformal coating to terminal box; install desiccant breather on motor cooling fan; conduct megohm resistance test ($\geq 100 \text{ M}\Omega$) every 6 months per NFPA 70B

SECTION 4 - INVENTORY DATA

- **Current Stock**: 1 complete pump assembly; 3 bearing cartridge kits; 4 mechanical seal kits; 2 impeller assemblies; 1 motor winding spare
- **Reorder Threshold**: Maintain minimum 1 complete unit + 2 bearing kits + 2 seal kits; trigger order when bearing kit stock reaches 1 unit
- **Supplier**: FluidDynamics Pump Corporation (primary); backup: Apex Marine Pumps Inc. (8-week lead time); bearing/seal kits available from multiple distributors (2–3 week lead time)

- **Lead Time & Logistics:** Complete pump assembly 6–8 weeks; bearing/seal kits 2–3 weeks; expedited air freight available for critical components (5-day delivery, 25% premium)

COMPLIANCE SUMMARY

Regulatory Alignment:

- All procedures comply with OSHA 1910.119 (Process Safety Management) and OSHA 1910.147 (Lockout/Tagout)
- Maintenance schedules follow API RP 54 (Recommended Practices for Offshore Drilling Fluid Operations)
- Pressure testing per API RP 579 (Fitness-for-Service)
- Vibration monitoring per ISO 20816 (Mechanical Vibration)
- Electrical safety per NFPA 70 (National Electrical Code)

Document Approval:

- Field Engineer: James R. Hutchins (MPO-ENG-4782)
- Date: 2024-01-18
- Next Review: 2025-01-18

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